FORMAL UNRAMIFIED, ETALE AND SMOOTH PROPERTIES CANCEL ON THE LEFT OF A FORMALLY UNRAMIFIED TRANSFORMATION

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is the whole lemma environment lines 2665-2676, including the three enumerated cases (1) unramified, (2) etale, (3) smooth.

Editorial note. Proof span is the complete author proof, lines 2678-2691; the closing sentence 'From this the lemma follows formally.' is retained verbatim and is the author's own reduction, not a gap in the retained argument.

Editorial note. No external_references are needed: the proof cites no label and no bibliography entry, and the statement cites nothing.

Editorial note. Preamble is the retained chapter preamble (sources/stacks-spaces-more-morphisms/retained-preamble.tex) copied mechanically from the native project preamble; it stops before the document body and contains no documentclass, so the assembler supplies the wrapper class.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads \usepackage[all]{xy}, and declaring xy again in extra_packages produces an option clash in the assembler wrapper. This matches the 303 sibling cards that reuse a retained or native preamble.

Editorial note. No figures or bibliography assets are referenced by this span.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Definitions of formally unramified, formally etale and formally smooth transformations of functors on (Sch/S)_fppf, and the square-zero thickening lifting criterion they are defined by. The correspondence argument in the retained author proof is not replaced by any cited result.

Lemma 0.1. *Let S be a scheme. Let $F, G, H : (Sch/S)^{opp}_{fppf} \to Sets$. Let $a : F \to G$, $b : G \to H$ be transformations of functors. Assume b is formally unramified.*

(1) *If $b \circ a$ is formally unramified then a is formally unramified.*
(2) *If $b \circ a$ is formally étale then a is formally étale.*
(3) *If $b \circ a$ is formally smooth then a is formally smooth.*

Proof. Let $T \subset T'$ be a closed immersion of affine schemes defined by an ideal of square zero. Let $g' : T' \to G$ and $f : T \to F$ be given such that $g'|_T = a \circ f$. Because b is formally unramified, there is a one to one correspondence between

$$\{f' : T' \to F \mid f = f'|_T \text{ and } a \circ f' = g'\}$$

and

$$\{f' : T' \to F \mid f = f'|_T \text{ and } b \circ a \circ f' = b \circ g'\}.$$

From this the lemma follows formally. □

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